Wireshark and OSI Model Practical Answers

Question 1. Install Wireshark, capture traffic while visiting a website, and note the protocol names shown in the packet list.

Answer: Wireshark is a packet analyzer used to inspect network traffic. The protocol names you see depend on your device, interface, browser, website, and whether the connection uses HTTP/2, HTTP/3, or other protocols. Follow these steps and record the actual protocols shown on your laptop.

Step-by-step procedure

1. Install: Download Wireshark from the official website, wireshark.org, and run the installer. On Windows, install Npcap if prompted because it enables packet capture. Use the official source and follow your institution's rules.

2. Start capture: Open Wireshark and identify the active interface (usually Wi-Fi or Ethernet) by observing which one shows activity. Double-click the interface to begin capturing.

3. Generate traffic: Open a browser and visit a website you are permitted to access, such as example.com or instagram.com. Wait for it to load; you do not need to log in or send private information.

4. Stop and inspect: Return to Wireshark and click the red square Stop button. Read the Protocol column. Select a packet and inspect the packet-details pane to see its protocol layers.

5. Record protocols: Write down the protocol names actually shown. Common examples may include DNS, TCP, TLS, QUIC, HTTP/2, HTTP/3, UDP, ARP, ICMP, IPv4/IPv6, Ethernet, or IEEE 802.11. You may not see all of these in every capture.

Example protocol table — guide only; use your actual results

Protocol	Common purpose
DNS	Looks up a domain name to find an IP address.
TCP	Provides reliable, ordered transport for many applications.
TLS	Protects application traffic using encryption and authentication.
QUIC	Transport over UDP used by HTTP/3 and other services.
HTTP / HTTP2 / HTTP3	Carries web requests and responses; version depends on the connection.
ARP or ICMP	ARP helps resolve local IPv4-to-MAC addresses; ICMP supports diagnostics.

Privacy note: Captures may contain IP addresses, device names, DNS lookups and other metadata. HTTPS usually encrypts web content, but not all connection metadata. Capture only on a device/network you are authorized to inspect, and avoid publicly sharing an unfiltered capture.

Question 2. Choose three packets from your capture and identify the OSI layer each packet mainly represents.

Answer: Use the Protocol and Info columns plus the packet-details pane. Real packets often contain several layers at once: an Ethernet frame can contain an IP packet, which contains TCP, which carries application data. The examples below show how to classify packets; replace the examples with rows that actually appear in your capture.

Example packet	Main OSI layer	How to recognize it
DNS query/response	Layer 7 — Application	Protocol shows DNS; details contain a query name or answer. DNS uses lower-layer protocols underne
TCP SYN or ACK	Layer 4 — Transport	Protocol shows TCP; details include source/destination ports and flags such as SYN or ACK.
IPv4/IPv6 packet	Layer 3 — Network	Details show source and destination IP addresses. An IP packet may also contain TCP or DNS data.

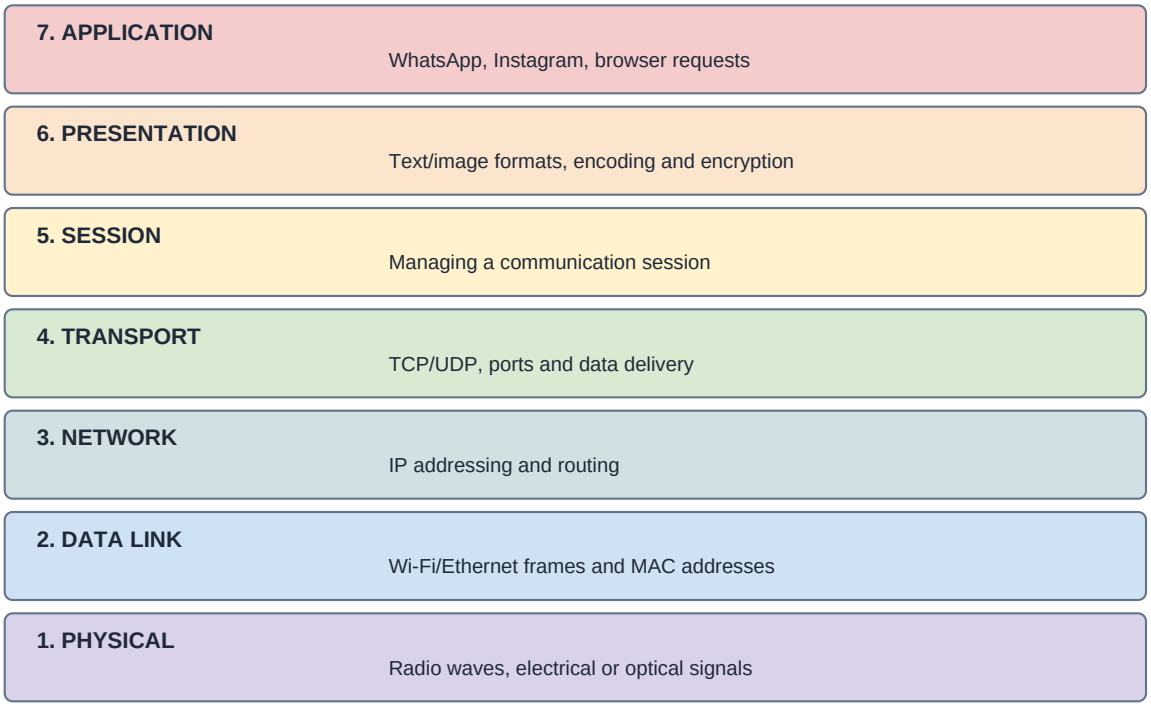
How to inspect packets

1. Click a row in the packet list.
2. Read the Protocol and Info columns.
3. Expand the relevant item in the packet-details pane, such as Domain Name System, Transmission Control Protocol, or Internet Protocol Version 4.
4. Record the protocol and its main OSI layer.
5. Remember that the OSI model is a learning framework; internet protocols do not always map perfectly to a single layer.

> **Important:** If the website uses QUIC/HTTP/3, you may see UDP and QUIC rather than TCP for that session. Choose packets that are present in your own capture; do not invent packet rows.

Question 3. Create a colorful OSI Model diagram and add a real-world example for each of the seven layers.

Answer: This colorful diagram separates the seven OSI layers. The examples are simplified because modern apps use several layers together.



Sender: data moves down; receiver: data moves up.

Layer	Purpose in simple words	Everyday example
7. Application	Network services used by applications.	WhatsApp messaging; browsing in Chrome.
6. Presentation	Data representation, encoding, compression and encryption concepts.	UTF-8 text, JPEG/PNG images, encryption concepts.
5. Session	Managing a logical communication session.	Maintaining an app or video-call session (simplified example).
4. Transport	Moves data between applications; uses ports.	TCP or UDP carrying app traffic.
3. Network	IP addressing and routing between networks.	IP packets travelling to a remote server.
2. Data Link	Local frames and link addressing.	Wi-Fi frames between phone and router.
1. Physical	Transmits bits as radio/electrical/optical signals.	Wi-Fi radio waves between phone and router.

Note: WhatsApp and Instagram do not belong to only one OSI layer. Each app uses multiple layers; the examples show where particular functions fit in the conceptual model.

Question 4. Explain how sending a WhatsApp message travels through at least four OSI layers.

Answer: When I type and send a WhatsApp message, the data passes down the sender's networking stack, travels over the network, and is processed upward on the receiving device. The following simplified explanation covers all seven layers.

Step 1 — Layer 7: Application

I open WhatsApp, type a message, and press Send. WhatsApp handles the message and prepares it for delivery through its service. The application layer is closest to the user and provides the network service used by the app.

Step 2 — Layer 6: Presentation

The message is represented in a form devices can process. Text is encoded, and communication is protected using encryption. In real internet stacks, encryption may be implemented at different layers, but it is often taught using the presentation-layer concept.

Step 3 — Layer 5: Session

The app maintains communication context so the exchange can continue. Session management is a useful conceptual description; modern protocols may handle these functions within the application or transport protocols.

Step 4 — Layer 4: Transport

Transport protocols carry application data between endpoints. Depending on the app's implementation and connection, traffic may use TCP or other transport mechanisms. Transport uses port numbers and may provide reliability, ordering, or congestion control.

Step 5 — Layer 3: Network

The data is carried in IP packets with source and destination IP addresses. Routers forward packets between networks toward the destination service. The path may cross several routers.

Step 6 — Layers 2 and 1: Data Link and Physical

On Wi-Fi, the phone sends frames over the wireless link to the home router. The data-link layer handles local frames and link addressing; the physical layer transmits bits as radio signals. The router forwards traffic through its ISP uplink.

Step 7 — Receiver processes data upward

The receiving device processes incoming signals and frames, checks IP and transport information, and passes protected application data to WhatsApp. The app then processes the message for display to the recipient.

Conclusion

The OSI model helps explain how data moves from an application through transport, network, data-link, and physical functions. Real protocols do not always match the seven-layer model exactly, but it is useful for understanding and troubleshooting communication.